

Jason Cabrejos

MS in Computer Science

BS in Mechanical Engineering

Minor in Computer Science

Grad date: May 2025

Focus: Robotics, Embedded, Automation, R&D

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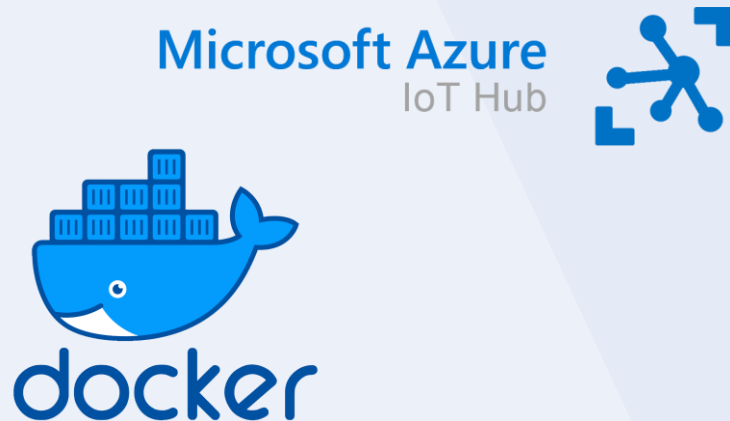
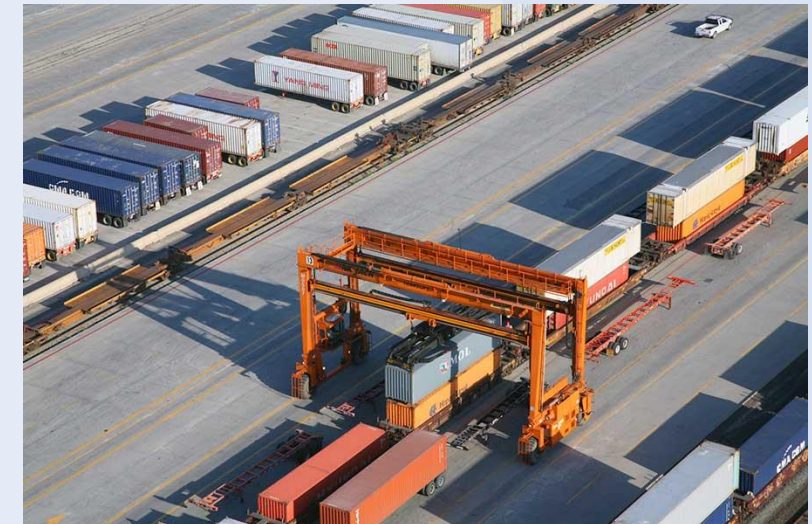
Table of Contents

Robotic Embedded Developer – BNSF.....	1
Researcher - UTA Research Institute.....	3
President - Rover Team.....	4
SHPE Competitions.....	6
Robotic Development Intern - Crown Castle.....	7
Researcher - The Laboratory of Advanced Materials Manufacturing and Analysis.....	8
CAD/PLM Intern - Beumer Group.....	9
Project Manager - Nasa L'Space.....	10
Structure Engineer – Aeromavs.....	11
Certifications.....	12

BNSF

Embedded Robotic Developer:

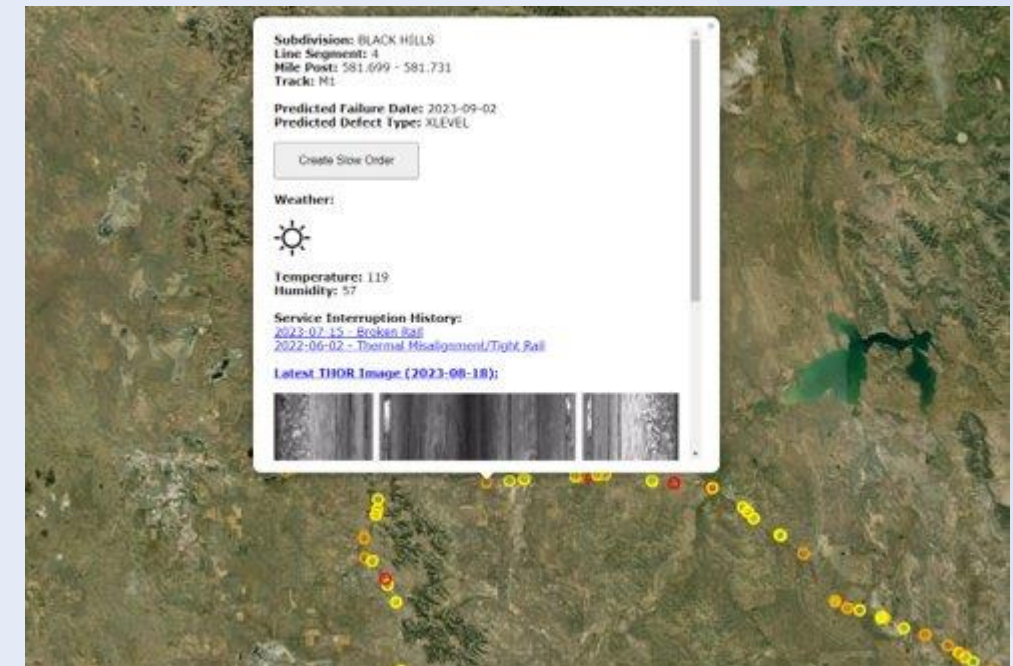
- Implemented a low-cost real-time inventory tracking system for under \$10K per unit merging technologies such as IoT, Networking, Power over Ethernet (PoE) systems, and edge computing solutions using the **Nvidia Jetson**
- Integrated GPS and a Machine Vision model to enable real-time inventory tracking during yard operations at a production scale at 85% accuracy
- Containerized using Docker and Azure IoT Hub to create a scalable and portable system for generating inventory maps



BNSF - Trackathon

Competition Winner 2023:

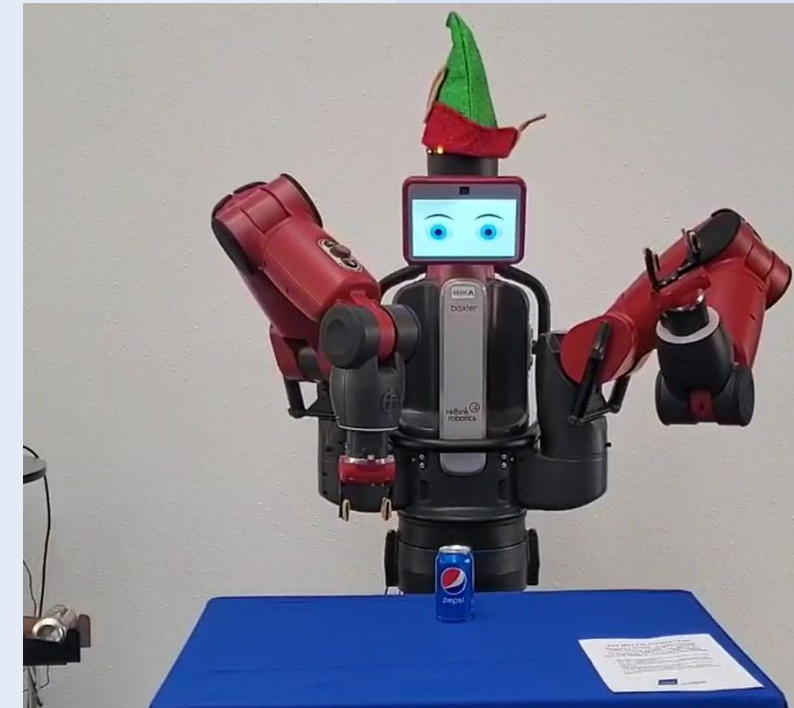
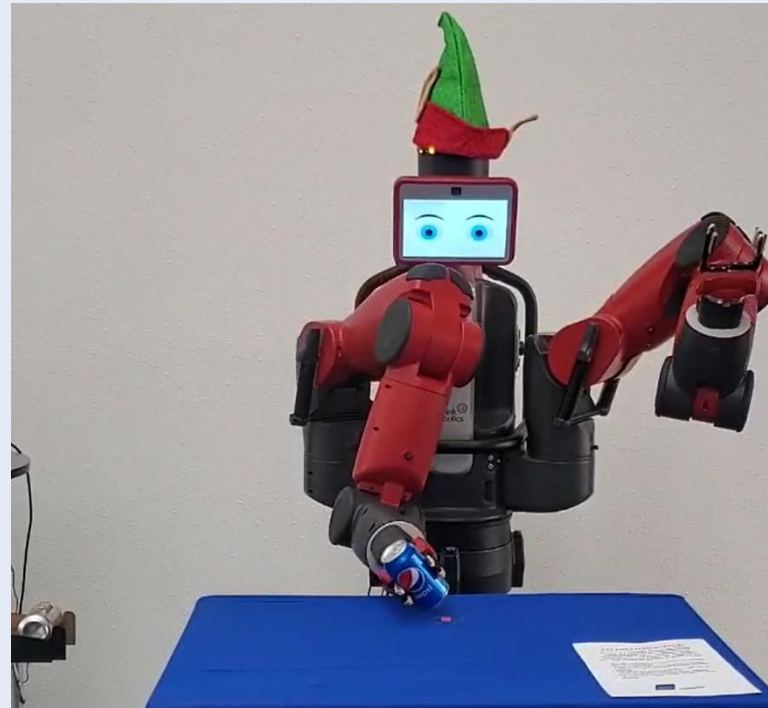
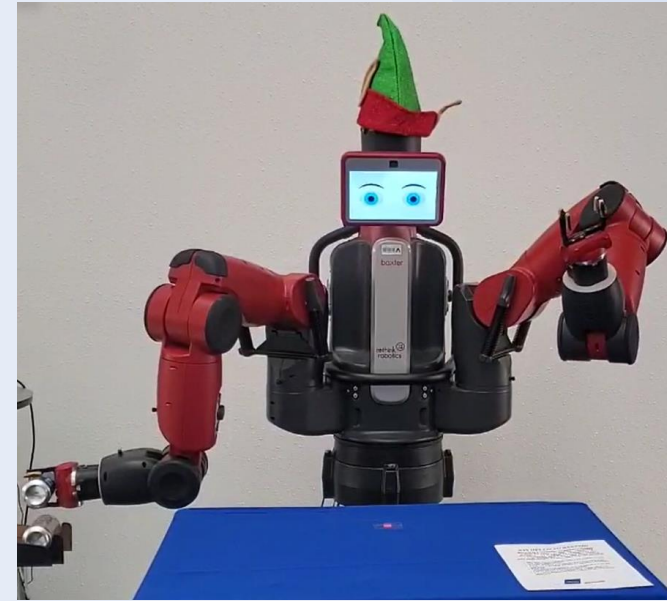
- Pitched and conceptualized a Machine Learning Model that predicts track health based on existing track maintenance data.
- The goal being to preemptively identify when repairs or inspections are needed, reducing Infrastructure-Related Service Interruptions.
- Created the financial pitch and project timeline for the product to come to market



UTA Research Institute

Research Assistant 2022:

- Worked on a various robotic systems such ReThink Robotics' Baxter and a road scanning robot
- Trained Baxter to recognize and retrieve soda cans using its cameras, robotic limbs, Robotic Operating System (ROS), and OpenCV
- Experience in Rapid Prototyping
 - 3d Printing
 - CNC Mill
 - Manual Mill and Lathe
 - CNC router
 - CNC Laser Cutter



Rover Team – Mechanical

Started as a member in 2017:

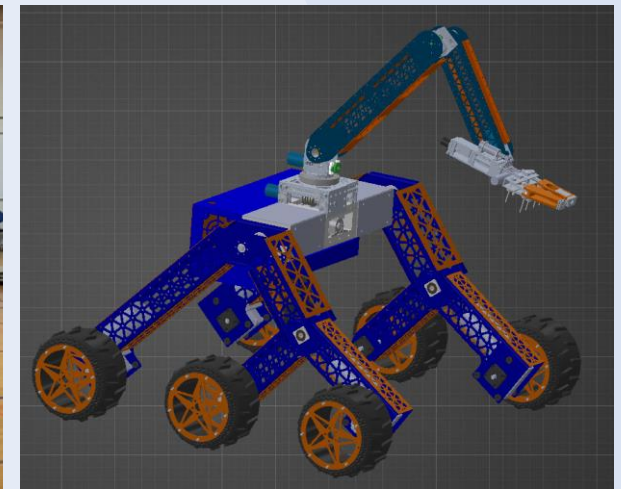
- Iterated robotic arm base joint on Solidworks.
- Soldered motors, motor controllers, and quick disconnects.
- Used Arduino to move and control the robotic arm

Arm and Effector Captain in 2018:

- Organized a team of 5 through a design cycle for the robotic arm.
- Designed a gripper capable of using a screwdriver
- Designed a sample return unit for taking soil samples and searching for signs of life

Rover Team President 2020-2021:

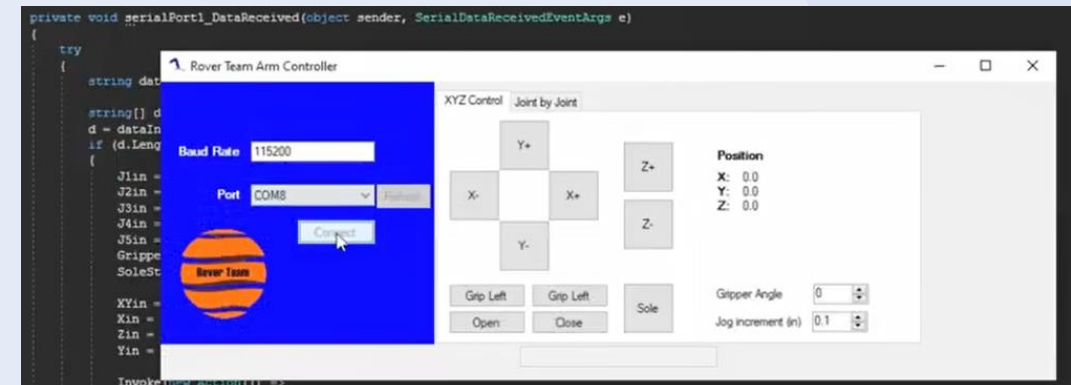
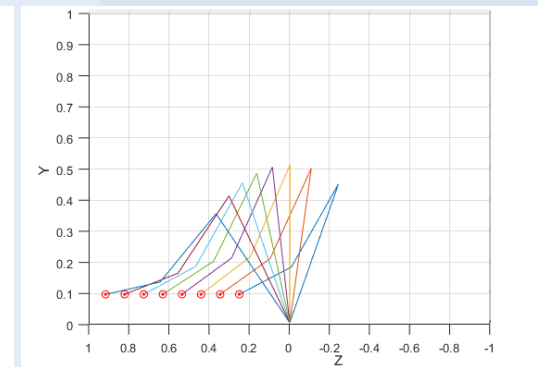
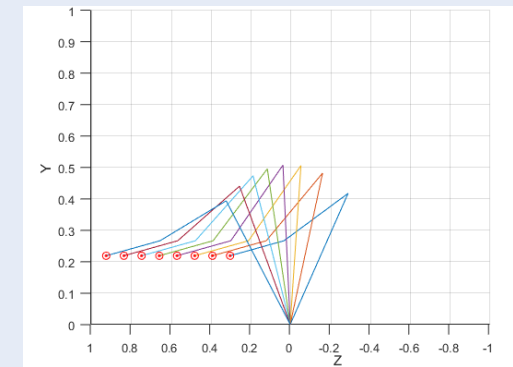
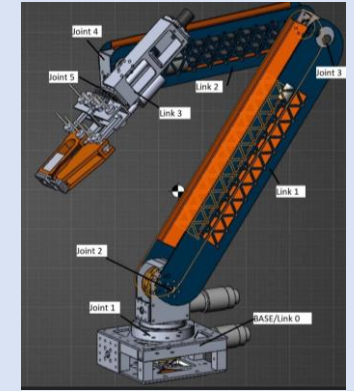
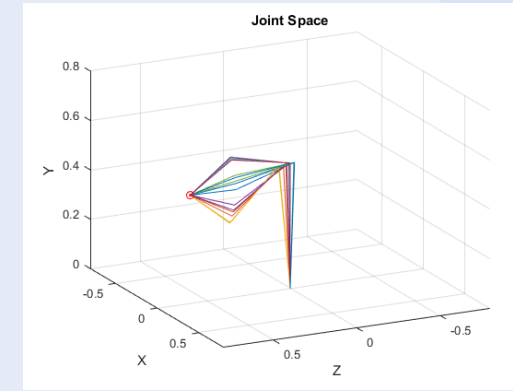
- Guided the team of 30 into a competition for the first time in 4 years.
- Advised 2 senior design teams into the development of improved tires and suspension.
- Finished 15th out of 88



Rover Team – Simulation

Rover Team President 2020-2021:

- Improved the robotic arm design to be easier to model dynamically
- Created a dynamic model with AutoLev and generated Equations of Motion for Joint Space and Operational Space Controllers (Sometimes called Task Space)
- Simulated robotic arm using Matlab to ensure it moved as intended and used the torque-speed curve for motor selection
- Created a Windows Forms program in Visual Studios to communicate with an Arduino and manually control the Arm by sending commands to the on-board Arduino



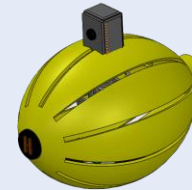
SHPE

Member (2021-Present)

- Nissan Design Challenge
 - Won 1st place Grand Prize
 - Selective application process, only 24 students were chosen to compete
 - Lead a team of 3 to pitch an idea to improve infant/child safety for Nissan/Infiniti Vehicles
 - Pitched a camera that would pick up excessive movement in the backseat and alert the driver to prevent them from constantly turning around to check on children, leading to safer driving
- Extreme Engineering Challenge
 - Tied for 1st Place in Semi-Finals
 - Out of 400 applicants, only 99 students were chosen to compete
 - Lead team of 11 to pitch a business concept for commercial space applications in Low Earth Orbit
 - Pitched a robot that allows earth customers an hourly rate to roam the Space Station and play games with other users.



 Eagle Eye



 Nu@rbit

Leading in Nu Space Solutions



Crown Castle - Robotic Development Intern



Robotic Development Intern 2022

- Extensive use of Ubuntu and Python on a ROS cube powered by an Nvidia Jetson
- A micro-trenching machine for fiber-optic cable deployment was improved upon by using Robotic Operating System (ROS) to read sensor data and automate operations to improve efficiency
 - Micro-trenching machine went from needing 6 to 4 people to operate once improvements were implemented
- CAN bus messages were tracked and recorded using an Ethernet Gateway.

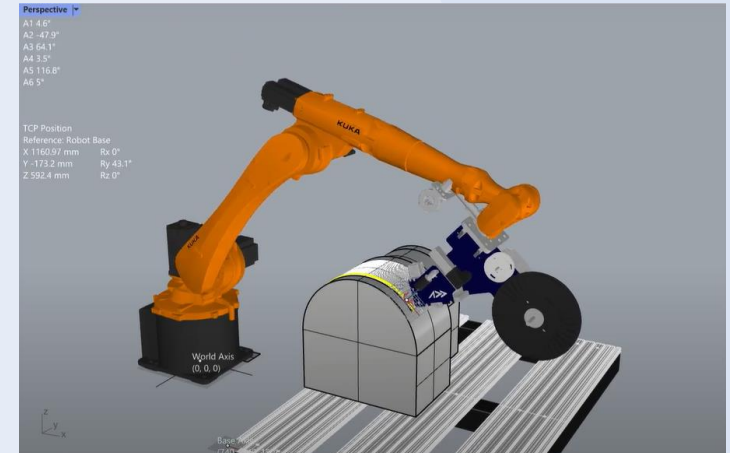
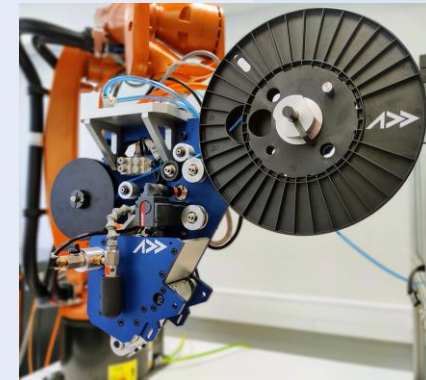
ROS



Laboratory of Advanced Materials Manufacturing and Analysis - Researcher

Researcher 2021

- Helped develop an Automatic Fiber Placement tool (AFP) to perform nonlinear and unique composite layups
- Configured Kuka Robotic Arm and the AFP tool using Programmable Logic Controllers (PLC's) though EtherNet/IP
- The research focuses on
 - Novel heating method used to improve consolidation between composite layers
 - Consider the force flow to create nonlinear composite layers. Leading to layers that follow the force flow's vector field
- Rhino CAD with the AddPath plugin were used to simulate toolpaths over different surface profiles



Beumer Group – CAD/PLM Intern

Intern 2019-2021

- C# in Visual Studio was used to create macros to simplify the design, drafting, and file sorting processes
- Used WinForms in Visual Studio to create user friendly GUI's

Macro examples:

- Copied the format and features of a master draft and applied to other similar drafts in the series
- Saved a copy of the pdf and sorted it based on part number in a shared folder
- Created user defined family of parts/assemblies from a master part/assembly
- Filled out information on PLM system based on part name
- Automatically dimensioned drafts to the company standard



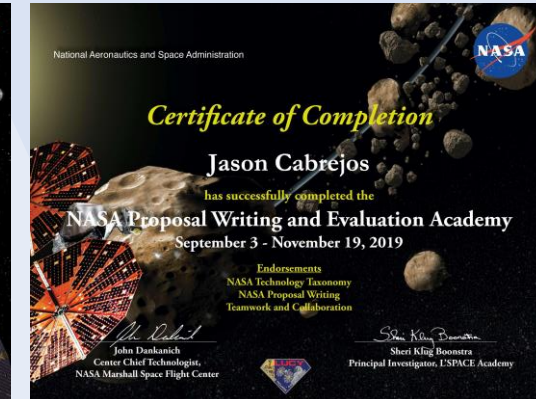
NASA L'Space

Mission Concept Academy – Deputy Project Manager 2019

- Conceptualized a submarine mission to one of the moons of Saturn – Titan
- Contacted Subject Matter Experts and learned how NASA operates and how they design a mission

Proposal Writing and Evaluation – Project Manager 2019

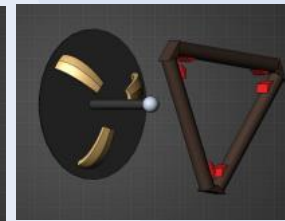
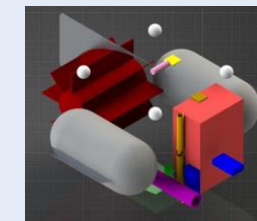
- Proposed an improvement for how the tool exchanges aligned in space on the end effectors of the Dextre and CanadaArm Robotic Manipulator system
- Purpose of the academy was to proposed a technological improvement/innovation



Tool Effector eXchange Array System (TEXAS)

PI: Jason Cabrejos & Amer Khalaf, Team 11

Goal / Objective - To minimize the need for emergency spacewalks by improving methods of detecting and dealing with potential failures and maintenance tasks. - By standardizing construction methods and improving a specialized end effector designed to have a diverse tool needed to make efficient and long-lasting repairs. - Improved version of Dextre's manipulators, tools, and interfaces.	
Team Overview - We have Mechanical, Electrical, Aerospace, and Computer Engineering and Science majors on our team. Some members have also been currently involved in robotics. 7.2.2 Maintenance System: Design and optimize specialized processes and tools that are required to perform maintenance during human exploration missions.	Metrics and Key Performance Parameters - The improved Dextre will reduce the number of parts, reduce cost, and make it easier to use for future astronauts on the gateway and other bases. - The improvements will have a positive impact on NASA schedules by reducing EVA to allow more time for onboard science. - Current Dextre Specs: Cost: \$20M, DOF: 15, Mass: 1710 Kg, Power to keep alive: 600 W



AeroMavs

Member in Rocketry - Structural Division (2017-2018)

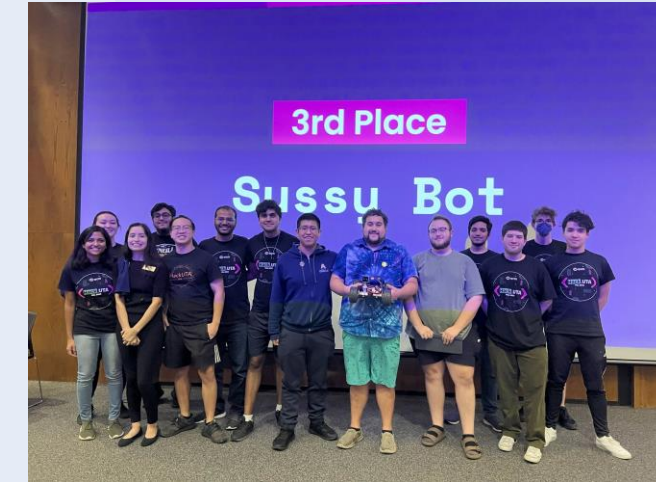
- Used OpenRocket to design cone, fin, and body
- Obtained L1 High Power Rocketry certification after successful launch and recovery of a model rocket
- Explored different fiberglass and epoxy combinations to optimize cost and strength of a rocket body
- Made carbon fiber composite fins and fiberglass composite body
- Competed in the Intercollegiate Rocket Engineering Competition (IREC)



Hackathons

Various Successes

- Trained a Tensorflow Computer Vision model to automate the recognition and sorting of waste using a robot– 3rd place
- Created an IoT system for agriculture irrigation with sensor and action nodes to count fruit, detect field health, and determine watering based on soil needs and local climate – 3rd place
- Presented a Machine Learning model and business plan to predict potential failures on the railway using big data – 1st place
- Improved child safety by pitching a camera system monitor activity without the driver needing to take their eyes of the road – 1st place

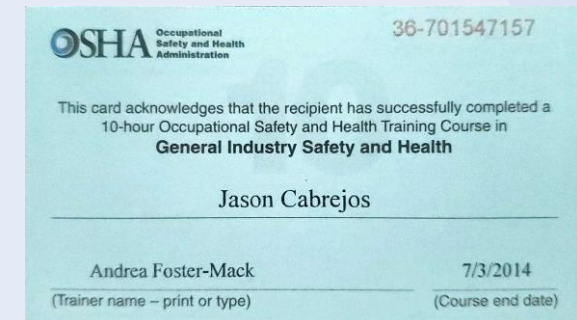


Certifications and Awards

Certified SOLIDWORKS Associate 2016

OSHA 10 hour General Industry Safety and Health

Deans List 2018





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